FARM MANAGEMENT SYSTEM

User Story:

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator,

I

want

to

use

a

centralized

Salesforce-based

platform

to

record

and

manage

farm,

crop,

farmer,

and

buyer

information

so

that

all

agricultural

data

is

stored

securely

and

systematically.

I

want

to

enter

and

update

farm

details,

crop

production

records,

and

buyer

transactions

easily

through

a

user-friendly

interface

with

proper

validation

rules

to

prevent

incorrect

entries.

I

also

want

the

system

to

generate

real-time

reports

and

interactive

dashboards

so

that

I

can

analyze

farm

productivity,

monitor

sales,

and

make

informed

decisions.

Additionally,

I

want

role-based

access

and

secure

data

storage

to

ensure

that

sensitive

information

is

protected

while

enabling

smooth

collaboration

among

users.

This

system

will

help

reduce

manual

paperwork,

minimize

errors,

improve

efficiency,

and

support

data-driven

farm

management.

1. INTRODUCTION 1.1 Project Overview :

The Farm Management System (FMS) is a Salesforce-based application designed to

digitize

and

streamline

agricultural

operations.

It

manages

data

related

to

farms,

crops,

farmers,

and

buyers

while

providing

real-time

insights

through

reports

and

dashboards.

The

system

helps

farmers

and

managers

monitor

agricultural

activities

efficiently

and

make

informed

decisions.

1.2 Purpose :

The main purpose of this project is to: ● Digitize agricultural data management ● Improve data accuracy and accessibility ● Automate record handling ● Enable real-time monitoring ● Support data-driven farming decisions ● Enhance productivity and transparency

2. IDEATION PHASE

2.1 Problem Statement (Customer Problem Statement Template)

PS I am (Customer)

I’m trying to

But Because Which makes me feel

PS-1 Farmer Maintain farm and crop records

Data is often incomplete or incorrect

Records are kept manually in registers

Frustrated and confused

PS-2 Farm Manager Monitor farm productivity and sales

Reports are delayed and inaccurate

Data is scattered across files and sheets

Stressed and inefficient

PS-3 Buyer Access crop availability and pricing

Information is not updated regularly

There is no centralized digital system

Uncertain and dissatisfied

PS-4 System Admin Manage user access and system security

Unauthorized access may occur

Access control is handled manually

Concerned and insecure

2.2 Empathy Map Canvas

Target Users:

● Farmers ● Farm Managers ● Buyers ● System Administrators

Empathy Analysis

● Think & Feel: Need accurate and timely farm data ● Hear: Feedback from agricultural officers ● See: Registers, paper files, spreadsheets ● Say & Do: Enter records manually ● Pain: Data loss, errors, delays ● Gain: Automation, fast reporting

2.3 Brainstorming

Brainstormed Ideas

● Manual farm records ● Excel-based management ● Web-based system ● Mobile farm app ● Salesforce-based FMS

Selected Idea

Salesforce-based Farm Management System.

3.REQUIREMENT ANALYSIS PHASE

3.1 Customer Journey Map

Stages:

1. Login 2. Access Farm Management App 3. Enter Farm and Crop Data 4. Register Farmers and Buyers 5. Generate Reports 6. Analyze Results 7. Decision Making

3.2 Solution Requirements

Functional Requirements

FR No	Requirement
FR-1	User Authentication
FR-2	Farm, Crop, Farmer, Buyer Management
FR-3	Data Entry and Validation
FR-4	Relationship Mapping
FR-5	Report Generation

FR-6 Dashboard Visualization

FR-7 User Role Management

Non-Functional Requirements

NFR Description Example

Usability Easy interface Simple Lightning UI

Security Data protection Profiles & roles

Reliability No data loss Cloud backup

Performance Fast access Reports in <3 sec

Availability 24/7 access Cloud hosting

Scalability Handles growth More farms/users

3.3 Data Flow Diagram

Data Flow:

User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) → Validation Rules

→

Relationships

→

Reports

→

Dashboards

3.4 Technology Stack

Layer Technology

Frontend Salesforce Lightning UI

Backend Salesforce Platform

Database Salesforce Cloud

Automation Validation Rules

Analytics Reports & Dashboards

4. PROJECT DESIGN PHASE

4.1 Problem–Solution Fit

The FMS replaces manual farming records with automated data management.

Cloud

storage

prevents

data

loss,

dashboards

improve

analysis,

and

real-time

reports

eliminate

delays.

This

ensures

accurate

and

timely

agricultural

insights.

4.2 Proposed Solution

Functional Requirements

Parameter Description

Problem Manual maintenance of farm, crop, farmer, and buyer records leading to errors, data loss, and delays.

Solution Salesforce-based Farm Management System with centralized data storage and automation.

Innovation Automated data entry validation, cloud storage, and real-time reporting dashboards.

Impact Improved data accuracy, faster reporting, and better decision-making for farmers and managers.es

Business Model Digital farm management and agricultural data services platform.

Scalability Supports multiple farms, users, and regions with cloud-based infrastructure.

4.3 Solution Architecture

Architecture Layers

1. Presentation Layer – Salesforce Lightning UI 2. Application Layer – Validation Rules, Relationships 3. Data Layer – Custom Objects 4. Analytics Layer – Reports and Dashboards

5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

● Created Salesforce Developer Org ● Developed custom objects ● Configured tabs and Lightning App ● Created fields and relationships ● Designed profiles and roles ● Created users and sharing rules ● Developed reports and dashboards ● Conducted testing and reviews

Product Backlog, Sprint Schedule, and Estimation (4 Marks) Use the below template to create product backlog and sprint schedule

Sprint Functional Requirement (Epic)

User Story Number

User Story / Task Story Points

Priority Team Members Sprint-1 Developer Setup

USN-1 As a system administrator, I want to create and configure a Salesforce developer account so that I can set up and deploy the Farm Management System.

3 High Member1

Sprint-2a Data Modeling USN-2 As an admin, I want to create Farm, Crop, Farmer, and Buyer objects with proper relationships so that agricultural data is stored and managed efficiently.

5 High Member2

Sprint-2 Data Modeling USN-3 As a staff member, I want tabs and a Lightning App for all farm modules so that I can easily navigate and perform daily operations.

5 High Member3

Sprint-3 Automation USN-4 As a system, I want to validate farm and crop details such as dates, quantities, and mandatory fields so that incorrect or incomplete records are prevented.

3 High Member4

Sprint-3 Automation USN-5 As a manager, I want automated flows for crop updates and buyer notifications so that farm operations are processed faster without manual intervention.

3 High Member4

Sprint-4 Security USN-6 As an administrator, I want role-based access control for farmers, managers, and buyers so that sensitive farm and transaction data is secure.

5 High Member2

Sprint-5 Reports USN-7 As a farm manager, I want reports showing crop yield, sales, and buyer transactions so that I can monitor farm performance and make informed decisions.

4 High Member3

Sprint-6 Dashboards USN-8 As a farm manager, I want dashboards showing crop productivity, sales trends, and operational statistics so that I can monitor performance visually.

4 Medium Member4

Project Tracker, Velocity & Burndown Chart: (4 Marks)

Sprint Total Story Points

Duration

Sprint Start Date

Sprint End Date (Planned)

Story Points Completed (as on Planned End Date)

Sprint Release Date (Actual)

Sprint-1 20 6 Days 24 Oct 2022 29 Oct 2022 20 29 Oct 2022 Sprint-2 20 6 Days 31 Oct 2022 05 Nov 2022 Sprint-3 20 6 Days 07 Nov 2022 12 Nov 2022 Sprint-4 20 6 Days 14 Nov 2022 19 Nov 2022

6. Project Development Phase: Project Description: The Farm Management System (FMS) in Salesforce is a comprehensive solution

designed

to

digitize

and

streamline

agricultural

operations.

This

application

empowers

farmers,

farm

managers,

and

stakeholders

by

managing

data

related

to

crops,

Farm,

Farmer,

and

buyers,

while

providing

actionable

insights

through

reports,

and

dashboards.

Milestone 1-Salesforce: What Is Salesforce? Salesforce is your customer success platform, designed to help you sell, service, market,

analyze,

and

connect

with

your

customers.

Salesforce has everything you need to run your business from anywhere. Using standard products and features, you can manage relationships with prospects and customers, collaborate and engage with employees and partners, and store your data securely in the

cloud.

Activity1: Creating a Salesforce Developer Org: A Developer org has all the features and licenses you need to get started with Salesforce. 1. Search Developer.salesforce.com/signup 2. Enter the following details like First name, last name, Email, Role, Company,

Country/Region,

Postal

code,

and

Username

must

be

unique.

3. Click sign me up, after a few min you will reserve a mail salesforce org and by using

the

verify

account

link

you

can

create

your

new

password.

4. Click save. 5. Search login.salesforce.com

6. By using username and password you can into the Salesforce org. The homepage will appear

below

Milestone 2-Object: What is an object? Salesforce objects are database tables that permit you to store data that is specific to an

organization.

It

consists

of

fields

(columns)

and

records

(rows).

In This Application We Use 4 Custom Objects: 1.Farm 2.Crop 3.Farmer 4.Buyer Activity 1: Create a custom object for Farm: To create a custom object, follow these steps: 1. From setup click on object manager. 2. Click create, select custom object.

3. Fill in the label as " Farm ". 4. Fill in the plural label as " Farms". 5. Record name: Farm Name" 6. Select the data type as "Text".

7. In the Optional Features section, select Allow Reports and Track Field History. 8. In the Deployment Status section, ensure Deployed is selected. 9. In the Search Status section, select Allow Search. 10. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout. 11. Leave everything else as is, and click Save. Activity 2: Creation of Crop object: 1. Click on the gear icon and then select Setup. 2. Click on the object manager tab just beside the home tab. 3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object. a. On the Custom Object Definition page, create the object as follows: b. Label: Crop c. Plural Label: Crops d. Record Name: Crop Name e. Select the data type as "Text". f. Check the Allow Reports checkbox g. Check the Allow Search checkbox h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout i. Click Save. Activity 3: Creation of Farmer object: 1. Click on the gear icon and then select Setup. 2. Click on the object manager tab just beside the home tab. 3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object. a. On the Custom Object Definition page, create the object as follows: b. Label: Farmer

c. Plural Label: Farmers d. Record Name: Farmer Name e. Select the data type as "Text". f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox. h. In the Object Creation Options section, select Add Notes and Attachments related list to default page layout. i. Click Save. Activity 4: Creation of Buyer object: 1. Click on the gear icon and then select Setup. 2. Click on the object manager tab just beside the home tab. 3. After the above steps, have a look on the extreme right you will find a Create Dropdown click on that and select Custom Object. a. On the Custom Object Definition page, create the object as follows: b. Label: Buyer c. Plural Label: Buyers d. Record Name: Buyer Name e. Select the data type as "Text". f. Check the Allow Reports checkbox. g. Check the Allow Search checkbox. h. In the Object Creation Options section, select Add Notes and Attachments related list

to

default

page

layout.

i. Click Save. Milestone 3 - Tab: Tab Tabs in Salesforce help users view the information at a glance. It displays the data of

objects

and

other

web

content

in

the

application.

Activity 1: Creation of Farm tab: Now create a custom tab. 1. Click on the Home tab, enter Tabs in Quick Find and select Tabs.

2. Under custom object tabs, click New.

3. For Object, select Farm.

4. For Tab Style, select any icon. 5. Leave all defaults as is. Click Next, Next, and Save. Activity 2: Creation of Crop tab: Now create a custom tab. 1. Click on Home tab, enter Tabs in Quick Find and select Tabs. 2. Under custom object tabs, click New. 3. For Object, select Crop. 4. For Tab Style, select any icon. 5. Leave all defaults as is. Click Next, Next, and Save. Activity 3: Creation of Farmer tab: Now create a custom tab. 1. Click on the Home tab, enter Tabs in Quick Find and select Tabs. 2. Under custom object tabs, click New. 3. For Object, select Farmer. 4. For Tab Style, select any icon. 5. Leave all defaults as is. Click Next, Next, and Save. Activity 4: Creation of Buyer tab: Now create a custom tab. 1. Click on Home tab, enter Tabs in Quick Find and select Tabs. 2. Under custom object tabs, click New. 3. For Object, select Buyer 4. For Tab Style, select any icon. 5. Leave all defaults as is. Click Next, Next, and Save. Milestone 4 - Lightning App:

What is an App? Apps in Salesforce area group of tabs that help the application function by working together

asa

unit.

It

has

a

name,a

logo,

and

a

particular

set

of

tabs.

The

simplest

app

usually

has

just

two

tabs.

Activity 1: Create the Farm Management System app

● From Setup, enter App Manager in the Quick Find and select App Manager. ● Click New Lightning App. Farm Management System as the App Name, then click

Next

● Under App Options, leave the default selections and click Next. ● Under Utility Items, leave as is and click Next. ● From Available Items, select Farm, Crop,Farmer,Buyer Reports,and Dashboards and move them to Selected Items. Click Next. ● From Available

Profiles,

select

System

Administrator

and

move

it

to

Selected

Profiles.Click

Save

&

Finish.

● To verify your changes, click the App Launcher, type Farm Management System and

select

the

Job

Application

app.

Note: 1. App Launcher-Displays available apps. 2. App Name-Displays the current selected app.

3. Navigation menu-Displays the tabs available inside the app.

Milestone 5 - Fields and Relationship: What are fields? Fields in Salesforce represent what the columns represent in relational databases. It can

store

data

values

which

are

required

for

a

particular

object

in

a

record.

Activity 1: Creation of fields for the Farm objects: 1. click the gear icon and select Setup. This launches Setup in a new tab. 2. Click the Object Manager tab next to Home. 3. Select Farm. 4. Select Fields & Relationships from the left navigation, and click New

5.Choose Data type as a picklist,Click next

6.Enter Field Label (Farm Type),and Enter Values as (Crop,Livestock,Mixed) 7.Next,Next Click Save.

Now let's create the other fields following Activity1 steps 1 to 4 and we must choose the

data

types

of

the

fields

carefully.

1.Next we Create another Picklist Field,Enter field label (Soil Quality) enter values

(Organic,Cover

Crops,Composting)

2.Again we Create Picklist field,Field Label (Irrigation type), Enter Values (Sprinkler

irrigation,Drip

irrigation,localized

irrigation)

3.Choose data type as a text,enter field label(Location) 4.Choose data type as text,enter

field

label

(size)

5.Create a Master Detail relationShip with Farmer Object. Activity 2: Creation of fields for the Crop objects: 1.Choose Data type As a text,Enter Field Label name (Crop Name),Next save. 2.Select

Data

type

as

date

,Enter

Field

label

(Planting

Date),then

save

.

3.Again we select data type as a date,Enter field labe (Harvest date),save 4.Next We Create Lookup relationship on crop object realted to Farm Object. 5.We select adata type as currency,Enter field Label (Actual Yield), Next Save it. 6.Select

Data type as a Number,enter Field Label (Total Acres Harvested),Enter

Maximum

Length(20)

and

save.

7.Now We create Formula Field For That Select data Type As a Formula,Click Next.

8.Enter Field Label (Expected Yield per Acre),Select formula return type as Currency,click

Next.

9.Enter a Formula Actual Yield Divide By Total Acre Harvested, then click next and save.

Activity 3: Creation of fields for the Farmer objects: 1.Select Data type Phone,Enter field label(Contact Number),next Save. 2.Select Data type Picklist,Enter Field Label(State),Enter values (Telangana,Andhra

Pradesh,Tamilnadu),save.

3.Choose Data type Picklist,Enter Field Label (District),Enter Values of district,next save. 4.Again select Data type as Picklist,Enter Field Label (Village),Enter Values of

villages,next

save.

Creation of field dependency : 1.Now create Field Dependency For District field,Open Object Manager,Choose Farmer

object

,select

Fields&Relationships,On

Right

side

We

have

Field

Dependency

option,select

that

option.

2.Click new, Select Controlling Field as State,Dependent Field As District,Click on

Continue.

3. Select the appropriate Value in each column by double-clicking them. 4. For Ex. Telangana - Yadadri Bhongir. 5.Click Include Values. And it is also the same for Andhra Pradesh, Tamilnadu with its

city.

6.Click Preview, then test the dependency by selecting a different State and viewing the

associate

Values

available

for

Particular

state.

7. Click Close to close the preview window, Click Save.

Activity 4: Creation of fields for the Buyer objects: 1.Select Data type as a Phone,Enter a Field Label(Contact),Save. 2.Select Data type as an Email,Enter Field Label(Email), Next save. 3.Select Data type Text area,Enter Field Label(Address),Save it. 4.Select Data type Picklist, Field Label(Buyer Type),Enter values

(Wholesale,Retail),Click

Save.

Milestone 6- Profile: What is a profile? A profile is a group/collection of settings and permissions that define what a user can do

in

salesforce. A profile controls “Object permissions, Field permissions, User permissions,

Tab

settings, App settings, Apex class access, Visualforce page access, Page layouts, Record

Types,

Login

hours

&

Login

IP

ranges.

A profile can be assigned to many users, but user can be assigned single profile at a time. Activity 1: Create a custom profile: 1. From setup , enter profiles in Quick Find box 2. Select profiles (Standard user). 3. Click clone. 4. For Profile, enter Sales Department.

5. Click save.

Activity 2: Create a custom profile-2: Create a profile with the profile name as “ Farm Manager” . 6. From setup , enter profiles in Quick Find box 7. Select profiles (Standard user). 8. Click clone. Milestone 7- Role In Salesforce, roles are used to determine which users have access to certain data and

functions

within

the

system.

They

are

also

used

to

define

the

reporting

hierarchy

within

an

organization.

Users

with

higher

roles

have

greater

access

to

data

and

more

control

over

the

system.

Activity-1 Creation of role 1. From the Quick find box search for the role and click on the roles option 2. select the set up roles option

3.Below the CEO click on add role and enter the label name as a ” Farm Manager ” and

role

name

will

be

Automatically

populated

and

click

on

save.

Milestone 8-User : What is a user? A user is anyone who logs into Salesforce. Users are employees at your company, such as

sales

reps,

managers,

and

IT

specialists,

who

need

access

to

the

company's

records.

Every

user

in

Salesforce has a user account. The user account identifies the user, and the user account

settings

determine

what

features

and

records

the

user

can

access.

NOTE- As Salesforce license can only be used by 2 Users at a time in Dev Org, so If you

don’t

find

salesforce license then deactivate a user who has salesforce license Or change the license

type

from

Salesforce

to

any

other.

Activity 1: To Create a user: 1. From Setup, enter Users in the Quick Find box, then select Users. 2. Click New User.

3. Enter First name as Farm and last name as Manager. 4. Enter the user’s name and email address and a unique username in the form of an email

address.

By

default,

the

username

is

the

same

as

the

email

address.

5. Select the role Farm Manager. 6. Select user License as Standard Platform User. 7. Select profile(Farm Manager). 8. Click save. Activity 2: Create a user with a username as “Sales Department”, and assign him the sales Manager

profile.

Follow

the

steps

from

above

Activity

Milestone 9-Sharing Rules: What are Sharing Rules? Sharing rules help users to share records based on conditions. It is basically created for

objects

whose

organization-wide

defaults

(OWD)

are

set

to

public

read-only

or

private

because

sharing

rules

can

only

extend

the

access

and

not

restrict

it.

Types of sharing rules, •Owner-based Sharing Rules •Criteria-based Sharing Rules Activity 1: Create a sharing rule 1) Goto Sharing Settings, which can be found under the Quick Find section. 2) Scroll down and find the candidate object where a sharing rule needs to be added, and

then

click

on

New

to

create

a

new

sharing

rule.

3) Add the label of the sharing rule you want to make. 4) Select your rule type based on the criteria. 5) Select the field can join immediately check field from the Crop object. 6) Select the Crop as equal and value is True. 7) And in selecting the users to share with the section select roles and in that select Farm

Manager.

8) And in the section of select the level of access for the users give the access

Read/Write.

9) And save the rule.

Activity 2: Create a Sharing Rule to Share the records of Buyer to Sales department with the Access

of

Read/Write.

Follow the steps from above Activity. Milestone-10: User Adoption Activity 1:Create a Record(Farmer) ● Click on App Launcher on left side of screen. ● Search Farm Management & click on it. ● Click on Farmer Tab. ● Click new and fill details & Save.

Milestone 11-Reports: What are Reports? A report is a list of records that meet the criteria you define. It's displayed in rows and columns, and can be filtered, grouped, or displayed in a graphical chart. Every report is

stored

in

a

folder.

Folders

can

be

public,

hidden,

or

shared,

and

can

be

set

to

read-only

or

read/write.

Activity 1: Create a report: Create a report that displays rating of the account and which has type and account name. 1) Click on app launcher search for reports. 2) Click on the new report and select the category has Farmer with Farms.

3) In the details section select the option start report. 4) In Outline Click on Column Select Which Field you want to display in the Report. 5) Click Group,Select Which Fields You Want to Group. 6) Save the report by giving label name(Farmer with Farms) and save the folder as a

public

folder

and

save

the

report.

Activity 2: Create a Report using the Objects Crops, Buyers. Follow the steps from above Activity. Milestone 12-Dashboard: Dashboards provide more insights than reports as they combine the data from many

reports

and

show

a

summarized

result.

Looking

at

many

reports

at

a

time

gives

the

flexibility

of

combining

the

results

from

them

quickly.

Also

summaries

in

dashboards

help

us

decide

on

action

plans

quicker.

The

dashboards

can

contain

charts,

graphs

and

Tabular

data.

Activity 1: Create a Dashboard: 1. Click the Dashboard tab. 2. Click New Dashboard. 3. Name the dashboard Farmers with farms and click Create. 4. Click +Component. 5. Select the Farmers with Farms and click Select. 6. Select the Gauge chart and click Add. 7. Click Save and then Done.

Milestone-13: View Report and Dashboard: Activity 1: Report ● Click on App Launcher on left side of screen. ● Search Farm Management System & click on it. ● Click on Reports Tab. ● Click on Farmers with Farms & see records

Activity 2: Dashboard • Click on App Launcher on left side of screen. • Search Farm Management system & click on it. • Click on Dashboard Tab. • Click Farmers With farms & see records.

7. FUNCTIONAL & PERFORMANCE TESTING During the testing phase, screenshots were captured from the Salesforce interface to verify

system

functionality.

The

following

were

tested:

● Creation of Farm, Crop, Farmer, and Buyer records ● Validation of input fields ● Relationship and dependency checks ● Profile and role access testing ● Sharing rule verification ● Report and dashboard generation

Performance testing ensured smooth system response and stability during multiple user

operations.

8. ADVANTAGES & DISADVANTAGES

Advantages

● Centralized data system ● Reduced paperwork ● Improved accuracy ● Better monitoring ● Real-time analytics

Disadvantages

● Requires internet connection ● Salesforce license dependency ● Limited offline access

9. Conclusion: The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

10. FUTURE SCOPE

● Mobile application integration ● AI-based crop prediction ● Weather data integration ● IoT sensor connectivity ● Government scheme integration ● Market price forecasting

11. APPENDIX

● Source Code: Salesforce Configuration and Metadata ● Dataset: Farm, Crop, Farmer, and Buyer Records ● Tools Used: Salesforce Developer Org

Thank You